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Research paper

Seasonally varying relationship between stem respiration, increment and carbon allocation of Norway spruce trees

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Stem respiration is an important component of an ecosystem's carbon budget. Beside environmental factors, it depends highly on tree energy demands for stem growth. Determination of the relationship between stem growth and stem respiration would help to reveal the response of stem respiration to changing climate, which is expected to substantially affect tree growth. Common measurement of stem radial increment does not record all aspects of stem growth processes, especially those connected with cell wall thickening; therefore, the relationship between stem respiration and stem radial increment may vary depending on the wood cell growth differentiation phase. This study presents results from measurements of stem respiration and increment carried out for seven growing seasons in a young Norway spruce forest. Moreover, rates of carbon allocation to stems were modeled for these years. Stem respiration was divided into maintenance (R_m) and growth respiration (R_g) based upon the mature tissue method. There was a close relationship between R_g and daily stem radial increment (dSRI), and this relationship differed before and after dSRI seasonal maximum, which was around 19 June. Before this date, R_g increased exponentially with dSRI, while after this date logarithmically. This is a result of later maxima of R_g and its slower decrease when compared with dSRI, which is connected with energy demands for cell wall thickening. R_g reached a maxima at the end of June or in July. The maximum of carbon allocation to stem peaked in late summer, when R_g mostly tended to decrease. The overall contribution of R_g to stem CO_2 efflux amounted to 46.9% for the growing period from May to September and 38.2% for the year as a whole. This study shows that further deeper analysis of *in situ* stem growth and stem respiration dynamics is greatly needed, especially with a focus on wood formation on a cell level.

Keywords: carbon allocation, CO_2 efflux, growth respiration, maintenance respiration, mature tissue method, *Picea abies*, SRI.

Introduction

Stem respiration is an important component of an ecosystem's carbon budget. It usually accounts for about 7–25% of total ecosystem respiration in forest ecosystems (Bolstad et al. 2004, Zha et al. 2004, 2007, Gaumont-Guay et al. 2006, Acosta et al. 2008, Tang et al. 2008, Khomik et al. 2010). This contribution to total ecosystem respiration can vary both over the year (Khomik et al. 2010) and between years (Bolstad et al. 2004, Zha et al. 2007), depending upon the amount and respiration

activity of stem woody tissues, environmental conditions and activity of such other ecosystem components such as soil or leaves.

Stem respiration is usually divided into two main components: maintenance respiration (R_m) and growth respiration (R_g) (Amthor 2000). R_m serves to maintain existing biomass such as, for example, turnover of proteins, nutrient uptake and ion fluxes (Amthor 1989). The magnitude of R_m depends on the amount of living biomass, and its temporal patterns are

driven mainly by temperature since it is directly related to the enzymatic processes (Ryan 1991). R_g is respiration that covers energy demands for production of new biomass (Ryan 1990, Gruber et al. 2009), primarily to adding proteins, lipids, cellulose, minerals, etc., in newly produced cells. It is therefore highly dependent on assimilate supply from photosynthesis leaves and their allocation to a certain organ (Chan et al. 2018). During the growing season, stem respiration consists of both components, while outside of this period, in late autumn, winter and early spring, it consists only of R_m (Ryan 1990, Lavigne et al. 1996). These two assumptions are applied in a common approach for separating R_m and R_g , which is termed the 'mature tissue method'. This method assumes that the total respiration from mature tissue (i.e., tissue that is no longer growing) may be attributed to the maintenance respiration component, and this can be used to estimate annual maintenance respiration rates as the relationship between R_m and temperature is constant over the year. Any respiration exceeding this maintenance rate during the growing season is then considered to be R_g (Lavigne and Ryan 1997, Stockfors and Linder 1998).

There have been studies describing the relationship between stem respiration and stem increment rate (Maier 2001, Ceschia et al. 2002, Zha et al. 2004, Chan et al. 2018). However, common measurement of stem radial increment does not record all aspects of stem growth processes, especially those connected with cell wall thickening, which starts later and takes longer than stem volumetric changes. Ceschia et al. (2002) found a positive correlation between R_g and carbon fixed in new tissues, which is related to formation of new structures. Because cell volumetric growth is rather attributed to water intake (Hölttä et al. 2010), it supports an idea that individual phases of stem growth vary in energy demands, which may affect the relationship between R_g and stem increment, and that the relationship does not have to be simply linear as observed by Stockfors and Linder (1998), Maier (2001) or Zha et al. (2004).

The onset of stem growth activity in the spring is driven by temperature in temperate and cold climate (Rossi et al. 2008). This temperature effect on plant development has been demonstrated, for example, using sums of effective temperatures, i.e., sums of surpluses of mean daily above a given threshold (Skálová et al. 2015). Warming up due to the changing climate is therefore expected to affect stem growth by its earlier beginning and also by prolonging into late autumn (Dose and Menzel 2004, Koch et al. 2005). Determination of the relationship between stem growth and stem respiration will help to reveal the response of both stem respiration and forest ecosystem to the changing climate.

In this study, we investigated the relationship between stem respiration and stem increment to determine how it varies over the season. The research is based on continuous stem CO_2 efflux measurements carried out for seven growing seasons

in a young Norway spruce forest. The study's aims were to determine seasonal patterns of R_m , R_g , stem increment and stem carbon allocation.

We hypothesized that there will be a weak overall relationship between R_g and daily stem increment and that this relationship will vary over the growing season, for which we divided the season into two parts—increment before and after stem increment culmination. We also hypothesized that the seasonal course of R_g will agree rather with carbon allocation into the stem segment and that R_g will culminate during maximum carbon allocation.

Materials and methods

Experimental site

The experimental Norway spruce (*Picea abies* (L.) Karst) forest stand is situated within the Experimental Ecological Study Site at Bílý Kříž (49°30' N, 18°32' E). The site is located in the Moravian-Silesian Beskydy Mountains in the Czech Republic at the altitude of 875 m above sea level on a 12.5° slope with a south-south-west exposure. The weather is characterized by a mean annual temperature of 6.8 °C and a mean annual total precipitation of 1265 mm as determined for the period 1998–2013. The experimental forest had been planted in 1981 using 4-year-old seedlings. Tree density decreased from 1664 trees ha^{-1} in 2005 to 1256 trees ha^{-1} in 2013. There were two snow break events during the 2005/2006 and 2011/2012 winters, which reduced the tree density of the stand. Mean tree height and diameter at breast height increased from 11.2 to 15.3 m and from 13.3 to 18.2 cm, respectively. A detailed yearly description is summarized in Darenova et al. (2018). The soil is predominantly a Haplic Podzol according to the FAO soil classification system. Otherwise, the forest has not been exposed to any disturbances.

Stem diameter increment

Stem diameter at breast height (DBH; cm) was monitored during the whole growing seasons in the years 2005–13 using dendrometer bands (EMS Brno, Czech Republic; accuracy of 0.01 cm) on 15 sample trees within the stand. The trees were chosen across different diameters to representatively characterize the stand. Changes in DBH were measured once per 7, 14 or 21 days (with higher frequency till June, middle frequency in July and August, and lowest frequency at the end of season). Mean tree-ring width increment from these 15 trees was calculated for each investigated year. Cumulative stem radial increment (SRI) was modeled using the Gompertz function (Rossi et al. 2003, Oberhuber et al. 2014) fitted onto measured data by the method of least squares as follows:

$$\text{SRI} = a \cdot e^{-e^{(b-c \cdot \text{DOY})}} \quad (1)$$

where SRI (μm) is the cumulative radial increment in tree-ring width on the particular day of year (DOY); a , b and c are coefficients of the Gompertz function (a is maximum cumulative tree-ring width in the particular year (1637–2880 μm), b (6.2–7.9) and c (0.035–0.045) are shape coefficients). The individual fits are shown in Figure S1 available as Supplementary Data at *Tree Physiology* Online. The Gompertz function was derived to obtain daily stem radial increment rate (dSRI; $\mu\text{m day}^{-1}$):

$$\text{dSRI} = a \cdot c \cdot e^{-e^{(b-c \cdot \text{DOY})}} \cdot e^{(b-c \cdot \text{DOY})} \quad (2)$$

where a , b and c are the coefficients of the Gompertz function from the previous equation.

Stem CO₂ efflux

Stem CO₂ efflux was measured on eight trees within an area of 30 m². Stem diameter at breast height of these trees ranged between 12 and 21 cm in the beginning of the experiment in 2005 and between 13 and 25 cm at the end of the experiment in 2013. Although stem CO₂ efflux was measured on different trees that stem increment, the trees were chosen within the same diameter range to characterize the stand as well as possible.

CO₂ efflux from the stem surface into the atmosphere was measured by the automated, closed dynamic (nonsteady-state, through-flow) SAMTOC system. The system consisted of eight chambers and control units for chamber closing, an infrared gas analyzer (Li-840, LI-COR, Lincoln, NE, USA), and a personal computer with control software and additional analog input and digital output hardware. The system is described in more detail in Darenova et al. (2018). Stem CO₂ efflux was measured sequentially in all eight chambers, and the whole cycle of measurements was done every 2 h. Measurements were carried out between May and October from 2005 to 2013. The years 2009 and 2012 were excluded from analyses of stem CO₂ efflux because a large amount of data was missing due to malfunction of the measurement system.

Additional measurements

Stem temperature was measured continuously at the same time as stem CO₂ efflux using thermometers (PT 100, Sensit, Rožnov pod Radhoštěm, Czech Republic) permanently installed approximately 5 cm below the chambers. We made a small L-shaped cut through the bark and inserted carefully the small flat thermometer just under the bark into the cambium layer.

Precipitation was continuously measured using a rain gage (MetOne 386, Met One Instruments, Inc., Grants Pass, OR, USA) placed on a tower above the stand.

Air temperature at the level of the average stand height was measured using an EMS33 sensor (EMS Brno, Czech Republic). For each year, the sum of effective temperatures (surplus of

mean daily temperatures > 5 °C) was calculated beginning from 1 February, and the day of year (DOY) when this sum reached 100 °C was determined.

Stem carbon allocation

Detailed analysis of wood structure and formation was carried out based on wood micro-core samples that were taken from six trees with DBH of 21–25 cm according to the methodology of Rossi et al. (2006) in 2014–17. Permanent microscopic samples (one per tree) had been prepared for (i) tracheid morphology and (ii) wood formation kinetics of developing cells according to the methodology of von Arx et al. (2016) and Cuny et al. (2014). Radial cell wall thickness (RCWT, μm) and radial cell diameter (RCD, μm) of fully developed tracheids at the end of growing season were measured using the image processing software ImageJ (Abràmoff et al. 2005) at three parallel radial lines per sample in the years 2014–17.

Additionally, the kinetics of wood formation, such as duration of tracheid diameter elongation (DTRD, days) and cell-wall thickening (DCWT, days), were analyzed according to Wodzicki (1971) from regular 14-day wood sampling carried out during the growing season in 2017. After that, each developed tracheid was characterized by the relative position in the tree-ring, DTRD and DCWT. The relationship among these variables was used for timing of wood formation also in the years 2014–16 because of similar intra-tree-ring density variation (Ratgeber et al. 2006) (Figure 1).

Wood density in the years 2014–17 (G , kg m^{-3}) was determined as morphometric density of individual tracheids (Ratgeber et al. 2006, Cuny et al. 2014) dependent on RCWT (μm), RCD (μm) and cell wall density (G_{CW} , 1450 kg m^{-3} (Dinwoodie 1981)) according to the following equation

$$G = \frac{2 \text{ RCWT}}{\text{RCD}} G_{\text{CW}} \quad (3)$$

The seasonal dynamic of biomass allocation into tracheid cell-wall (G) was assigned by the middle day between DTRD and DCWT termination for individual tracheids in the years 2014–17. The annual pattern of G was fitted with the Gompertz function and the result was used to simulate this relationship for years 2005–13.

Stem carbon allocation (SCA, $\mu\text{mol m}^{-2} \text{ s}^{-1}$) was calculated on the basis of Gompertz function fit of SRI, which determined growth of stem segment volume. The stem segment size was equal to that of stem CO₂ efflux measurement. Conversion of stem volume growth to SCA was done using the record of seasonal dynamic of wood density based on the wood cell properties. Finally, the carbon content 0.5 kg carbon kg^{-1} dry matter (Penman et al. 2003) was used.

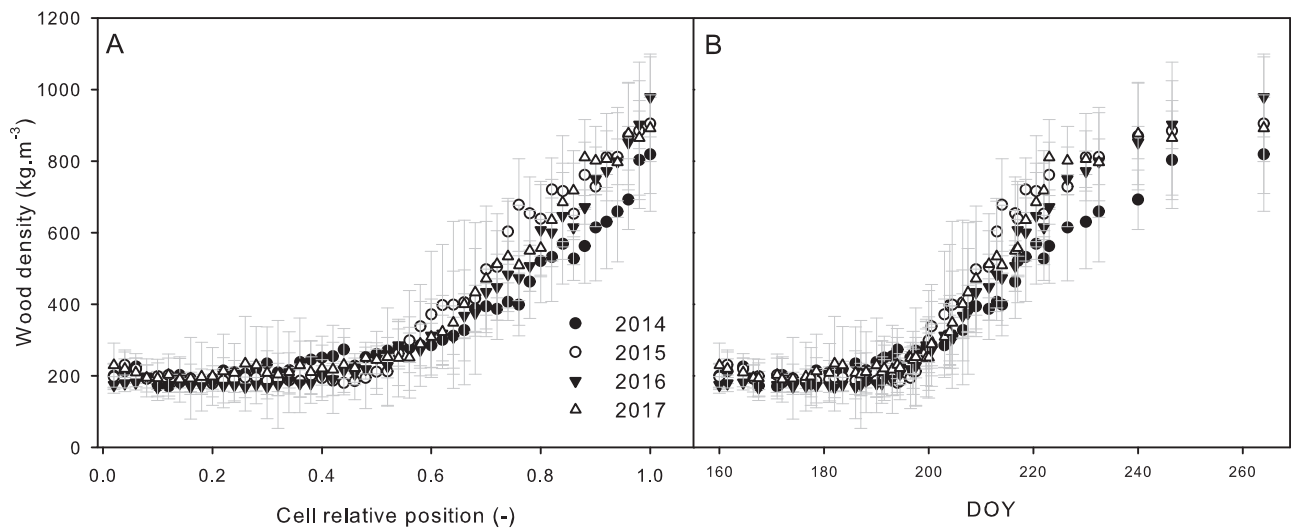


Figure 1. Wood density in relation to wood cell relative position a in an annual ring (A) and to day of year (B).

Data analyses

A substantial part of locally respired CO_2 can be transported upwards in xylem water (sap flow) during daytime (Teskey and McGuire 2007). As there is no sap flow during night time, we can say that night time CO_2 efflux is equal to stem respiration. Therefore, to limit the influence of this CO_2 transport in the xylem and to estimate stem respiration of the studied stem segment, night-time E_A (0:00–6:00 h, according to Darenova et al. 2019) was used to determine stem respiration (R_{st} , $\mu\text{molCO}_2 \text{ m}^{-2} \text{ s}^{-1}$) over whole 24-h periods according to the following equation:

$$R_{\text{st}} = \frac{R_{10\text{n}}}{Q_{10\text{n}}^{\frac{10 - T_{\text{st}}}{10}}} \quad (4)$$

Where $R_{10\text{n}}$ and $Q_{10\text{n}}$ were taken from Table 2 in Darenova et al. (2019) and represent night-time stem CO_2 efflux at 10°C and a coefficient that describes the change in E_A for a 10°C change in stem temperature (T_{st}), respectively.

To estimate maintenance respiration rates (R_m), mean Q_{10} and R_{10} were calculated over a continuous interval of 2 weeks during the second half of October over all experimental years ($R_{10\text{m}}$, and $Q_{10\text{m}}$, respectively). This was the latest period of measurements of stem CO_2 efflux, and no stem increment and stem carbon allocation were recorded during this time. Daily R_m was then estimated according to Eq. (4) just using parameters $R_{10\text{m}}$ and $Q_{10\text{m}}$, T_{st} being daily mean temperature.

Growth respiration (R_g) was then estimated as the difference between R_m and R_{st} :

$$R_g = R_{\text{st}} - R_m \quad (5)$$

To calculate annual sums of carbon respired by the studied stem segment, missing measurements outside of the growing seasons were calculated like R_m because we consider that R_{st} during these periods consisted only of this component.

The relationship between dSRI and R_g was evaluated separately for period before and after DOY 170, which was the mean day of maximum dSRI from all years. The best fit for the relationship between R_g and dSRI before DOY 170 was an exponential function.

$$y = a \cdot e^{b \cdot x} \quad (6)$$

and a logarithmic function after DOY 170

$$y = a \cdot \ln(x) + y_0 \quad (7)$$

The relationship between R_g and carbon allocation was fitted by a two segment piecewise linear fit

$$f = \begin{cases} \frac{y_1(T_1 - t) + y_2(t - t_1)}{T_1 - t_1}, & t_1 \leq t \leq T_1 \\ \frac{y_2(t_2 - t) + y_3(t - T_1)}{t_2 - T_1}, & T_1 \leq t \leq t_2 \end{cases} \quad (8)$$

where t_1 , T and t_2 are the time points representing the minimum, break point and maximum of the x-variable, respectively.

To assess the time lag of growth respiration behind stem radial increment and carbon allocation rate behind growth respiration, we plotted cumulative R_g against cumulative stem radial increment fitted by the modified Gompertz functions (Eq. (1)) cumulative R_g and cumulative SCA against cumulative R_g , respectively. Gradually, the time lag was increased by 1 day to find the best linear fit.

Correlations were analyzed using Pearson correlation test. All regression and correlation analyses were conducted using SigmaPlot 11.0 analytical software (Systat Software, San Jose, CA, USA).

Table 1. Mean annual radial stem increment, day of year when sum of effective temperatures (T_{ef}) reached 100 °C and dates when growth respiration (R_g) exceeded 50% ($R_g > 50\%$) and dropped below 50% ($R_g < 50\%$) of total stem respiration; seasonal (1 May–30 September) and annual (1 January–31 December) total respiration (R_{st}), maintenance respiration (R_m) and growth respiration (gC per m² of the stem surface) of spruce stems, and proportional contribution of R_g to R_{st} (% R_g).

	Radial stem increment (mm)	$T_{\text{ef}} = 100\text{ °C}$	$R_g > 50\%$	$R_g < 50\%$	R_{st}		R_m		R_g	% R_g	
					May–Sep	Jan–Dec	May–Sep	Jan–Dec		May–Sep	Jan–Dec
2005	2.8	122	162	338	322.7	364.4	132.8	174.6	189.8	58.8	52.1
2006	1.8	129	168	201	269.8	315.7	153.0	198.9	116.8	43.3	37.0
2007	2.1	117	129	200	277.7	328.3	145.2	195.7	132.6	47.7	40.4
2008	1.9	127	160	254	310.4	366.1	134.7	190.4	175.7	56.6	48.0
2010	1.8	121	168	267	268.9	315.5	131.4	178.0	137.5	51.1	43.6
2011	2.1	115	162	236	286.6	337.4	140.6	191.4	146.0	50.9	43.3
2013	1.7	119	171	267	297.1	343.3	143.9	190.4	153.3	51.6	44.6
Mean ($\pm$ SD)			162 ($\pm$ 7 days)	243 ($\pm$ 32 days)	290.5 ($\pm$ 20.6)	338.7 ($\pm$ 20.9)	140.2 ($\pm$ 7.8)	188.5 ($\pm$ 8.9)	150.2 ($\pm$ 25.3)	51.4 ($\pm$ 5.2)	44.1 ($\pm$ 4.9)

Results

Micrometeorological conditions

Micrometeorological characteristics of the individual seasons are summarized in detail in Darenova et al. (2018) indicating that season 2006 was the warmest and 2010 the coldest. The seasonal peak of air temperature was usually in July and August. The highest annual precipitation was recorded in 2010, which mainly was due to heavy rains in May. The driest season was in 2013, when there was an extremely dry summer, and in July and August, the monthly sums of precipitation did not exceed 50 mm (see Figure S2 available as Supplementary Data at *Tree Physiology Online*). The sum of effective temperatures above 5 °C (T_{ef}) exceeded 100 °C between 25 April and 9 May (DOY 115 and 129). This threshold was exceeded earliest in 2011 and the latest in 2006 (Table 1).

Stem radial increment and carbon allocation

Annual stem radial increment at breast height in the studied years ranged between 1.7 and 2.8 mm. The increment in 2005 greatly exceeded those in other years (Table 1). The tree stems started to grow during the first half of May. Maximum dSRI occurred between 14 June (in 2007; DOY 165) and 25 June (in 2006; DOY 176) (Figure 2) and ranged between 24.3 and 42.7 $\mu\text{m day}^{-1}$. The dSRI then gradually decreased until October.

The DOY when dSRI reached 1 $\mu\text{m day}^{-1}$ ranged between 121 and 135 (i.e., 1 May and 15 May). T_{ef} on these particular days ranged between 76.5 °C and 147.9 °C (Figure 3A). In analyzing the relationship between DOY when T_{ef} exceeded 100 °C and DOY when dSRI reached 1 $\mu\text{m day}^{-1}$, we found a significant correlation ($P = 0.005$) close to the 1:1 line (Figure 3B).

The seasonal pattern of modeled carbon allocation to the stem segment showed a slow increase during May and June, and following acceleration peaking around DOY 216 (4 August). During this peak, the carbon allocation rate reached 16.3–28.6 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Figure 2). The peak of carbon allocation

rate occurred between 40 and 51 days later than that of dSRI (Table 2).

Stem CO_2 efflux

Mean daily R_{st} ranged between 0.21 and 5.30 $\mu\text{mol m}^{-2} \text{s}^{-1}$ between 1 May and 30 September (Figure 4) with maxima between June and August. In 2006 and 2013, R_{st} decreased rapidly in July and August, respectively, and remained below 2 $\mu\text{mol m}^{-2} \text{s}^{-1}$ despite comparable temperatures as in other years.

The parameters used for calculating R_m were the following: $R_{10} = 0.56 \mu\text{mol m}^{-2} \text{s}^{-1}$ and $Q_{10} = 2.60$. Mean daily R_m , and R_g were in the ranges 0.21–2.57 and 0.00–3.13 $\mu\text{mol m}^{-2} \text{s}^{-1}$, respectively, between 1 May and 30 September (Figure 4). During May, R_m was the main component of stem respiration. At the end of May or in June, however, the contribution of R_g to R_{st} increased rapidly. R_g started to exceed R_m between 29 May and 20 June, as summarized in Table 1 for individual years. R_g became lower than R_m again between 19 July and 4 October.

Stem respiration vs stem increment rate and carbon allocation

An analysis of the relationships between dSRI and 7-day running average of R_g from all studied seasons showed that R_g generally increased with the SRI rate but that the relationship between the two differed between the periods of stem growth acceleration (DOY <170) versus deceleration (DOY >170) (Figure 5). From the beginning of the growing season until the culmination of dSRI (between May and the middle of June), the relationship was described with an exponential function. Stem radial increment rate rapidly increased from the onset of cambial activity while the increase in R_g was gradual at the beginning. In May, the change of dSRI from 0 to 20 $\mu\text{m day}^{-1}$ was followed by R_g increase only from 0.1 to 0.3 $\mu\text{mol m}^{-2} \text{s}^{-1}$. From the middle of June until the end of the growing season, the relationship between dSRI and R_g became logarithmic (Figure 5). First, dSRI declined rapidly from 30 to 10 $\mu\text{m day}^{-1}$ until the middle of August even as R_g changed only minimally (from 1.75 to

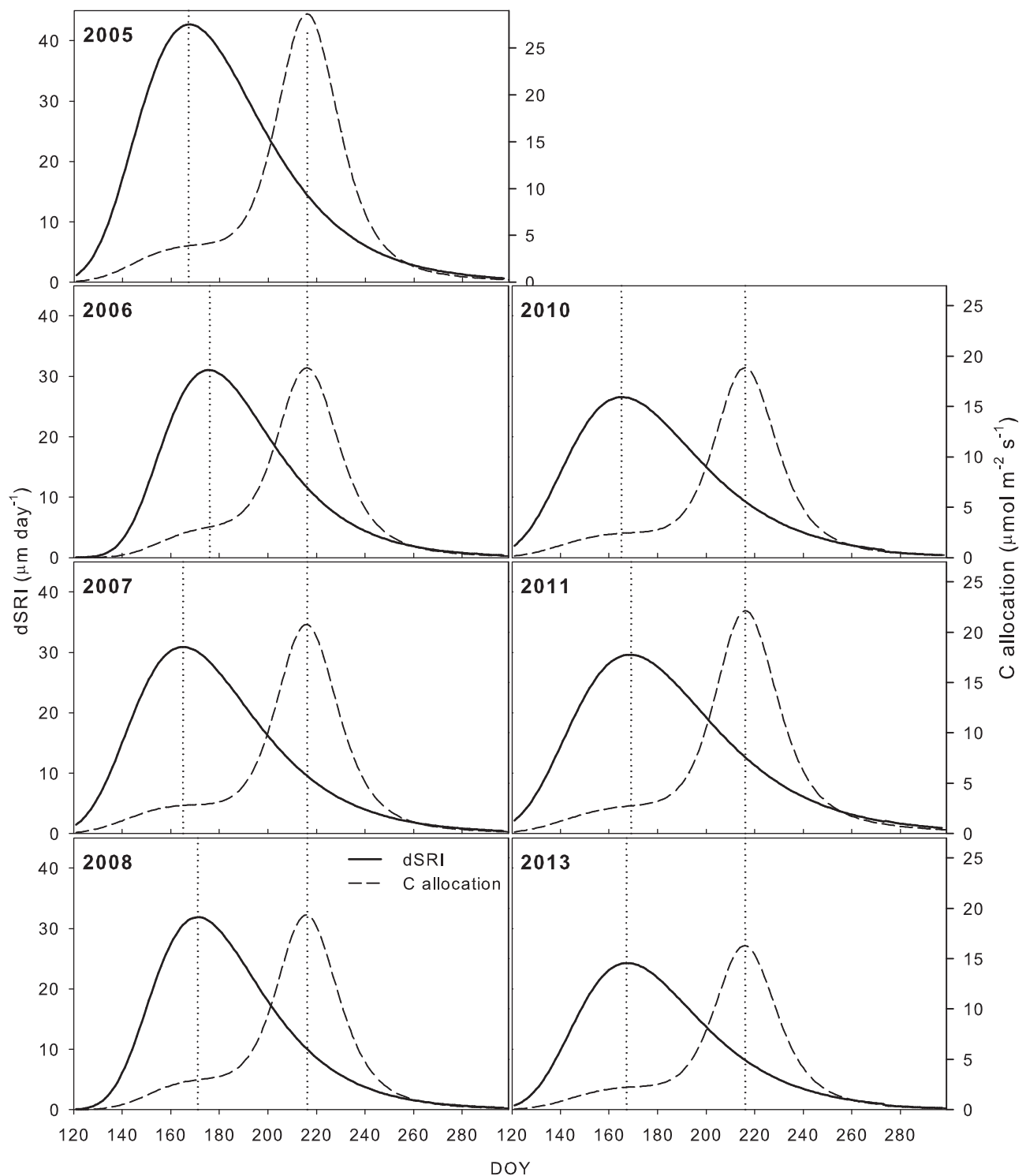


Figure 2. Daily stem radial increment (dSRI) and carbon (C) allocation to tree stem during seven studied seasons.

$1.25 \mu\text{mol m}^{-2} \text{s}^{-1}$). A substantial decrease in R_g was then observed between the middle of August and the beginning of October. The time lag of growth respiration behind SRI amounted mostly around 35 days (Table 2). The exceptions were years 2006 and 2007 when the time lag was only 9 and 14 days, respectively.

The highest rates of carbon allocation to stem were observed in July and August (Figure 2) when R_g mostly tended to decrease (Figure 4). The piecewise regression between carbon allocation and R_g was found to be statistically significant ($R^2 = 0.70$; $P < 0.0001$) (Figure 5). Before the carbon allocation maximum (DOY 216), regression coefficient increased

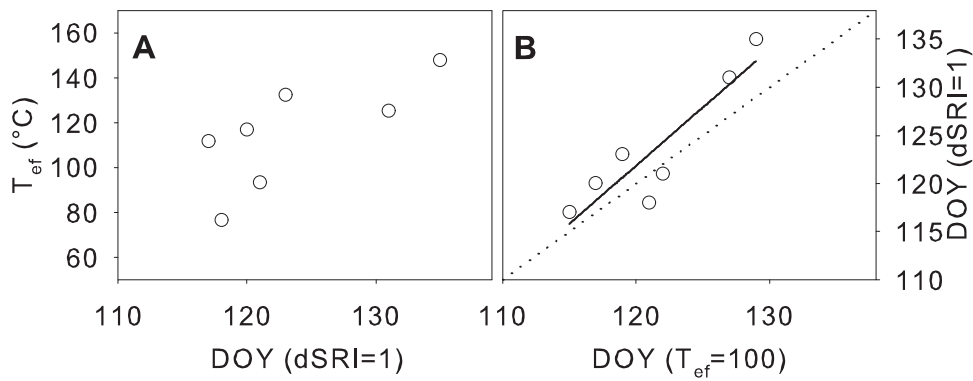


Figure 3. (A) Sums of effective temperatures above 5 °C (T_{ef}) on day of year (DOY) when daily stem radial increment (dSRI) reached 1 $\mu\text{m day}^{-1}$ in individual years and (B) Relationship between DOY when dSRI reached 1 $\mu\text{m day}^{-1}$ and DOY when T_{ef} reached 100 °C (dotted line represents 1:1).

Table 2. Time lag (days) of stem carbon allocation (SCA) behind daily stem radial increment (dSRI), of growth respiration (R_g) behind dSRI and of SCA behind R_g .

	2005	2006	2007	2008	2010	2011	2011
dSRI – SCA	49	40	51	45	51	47	49
dSRI – R_g	35	9	14	31	38	36	33
R_g – SCA	5	25	29	7	7	7	8

to 0.77 ($P < 0.0001$), while after R^2 was only 0.64, but the regression was still significant ($P < 0.0041$). R_g increased fast and tightly with carbon allocation, up to carbon allocation of 4.4 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Then no increase was observed, and the relationship was weak. The best fit between cumulative R_g and cumulative carbon allocation was found for the time lag of 29 days in 2007, 25 days in 2006 and for 5–8 days in other years (Table 2).

Discussion

Stem radial increment and carbon allocation

Previous studies on Norway spruce confirmed that wood production (measured as stem radial increment) takes place usually from April to October, depending upon site conditions (Rossi et al. 2008, Cuny et al. 2012). Giagli et al. (2016) studied cambial activity and xylogenesis in Norway spruce at a similar site in the Czech Republic between 1981 and 1989. Those authors found that cambium was activated in the beginning of May (similarly to our study) and that the maximum xylem cell production occurred between 8 June and 6 July. This range includes also the maximum stem increment period observed in our study (15–25 June).

The driving factor for activating cambium and initiating stem growth in spring is air temperature. Other factors, such as for example water availability, affect wood formation later in the growing season (Horáček et al. 1999, Gričar et al. 2007). Similarly, the effect has been studied of temperature prior to the growing season on bud burst, which is closely followed by stem growth (Chalupa 1969). Hannerz (1999) found that a model

based on sums of temperature above the threshold of 5 °C had the best results for estimating the beginning of bud burst. In our study, the sums of effective temperatures above 5 °C (T_{ef}) on the day when dSRI reached 0.1 $\mu\text{m day}^{-1}$ had quite broad range (76.5–147.9 °C), and this might lead to an assumption that start of stem growth is not driven by temperature. However, a strong correlation between DOY when T_{ef} exceeded 100 °C and DOY when dSRI reached 1 $\mu\text{m day}^{-1}$ rather contradicts any such hypothesis. Determination of stem growth response to temperature increase in the spring allows estimating the onset of stem growth in the future when the earlier temperature increase after winter is expected.

Changes in stem diameter result from both irreversible radial expansion of the stem as a consequence of the development of newly formed tissues (Cuny et al. 2015), and reversible radial stem shrinkage and expansion through changing water contents in stem tissues (De Swaef et al. 2015). The latter follows a diurnal pattern with a minimum stem radius in the afternoon and a maximum stem radius in the early morning. Trees do not have to fully recover during night time in dry periods; therefore, stem tissues are not completely refilled and the stem remains in a partly shrunk state (Kozłowski and Winget 1964, Zweifel et al. 2001). Dendrometer bands used in our study were not able to distinguish between the irreversible growth and the changes due to water status. We assume that short term fluctuations in water status were suppressed by fitting the data by the Gompertz function. We believe that this approach provided a real seasonal pattern of the stem growth.

While stem radial increment increased rapidly in the beginning of the growing season peaking in June, carbon allocation into

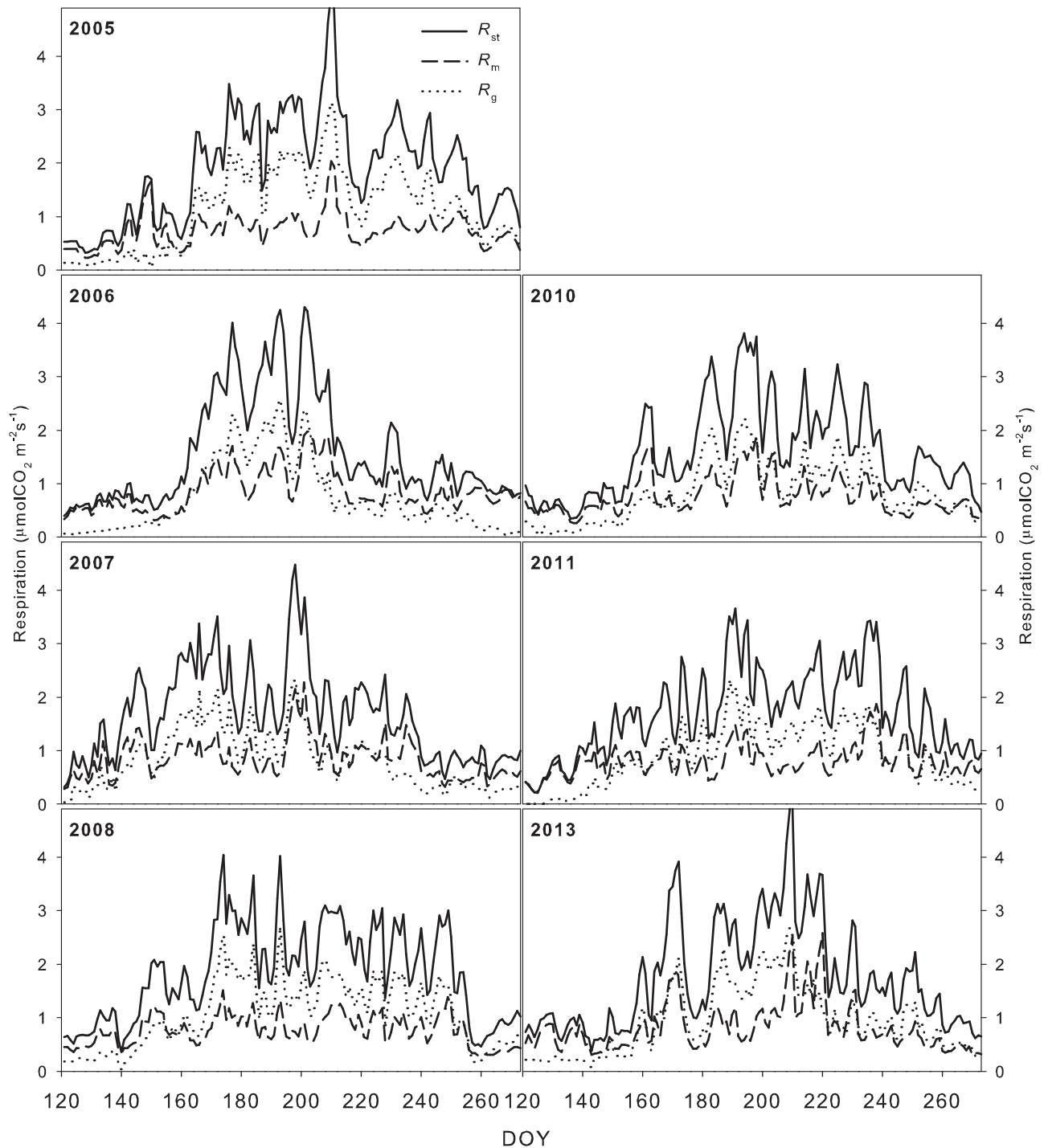


Figure 4. Mean daily total stem respiration (R_{st}), maintenance respiration (R_m) and growth respiration (R_g) in spruce trees.

stem increased slowly during this period and peaked in August. Such a time shift was also observed by [Cuny et al. \(2015\)](#) who observed 40-day time lag of woody tissue production ($\text{gC day}^{-1} \text{ tree}^{-1}$) behind maximum xylem size increase ($\mu\text{m day}^{-1}$). That is only a maximum 11 days shorter than the lag of maximum allocation behind the maximum stem increment rate observed in our study (the time lag was 40–51 days). In early stages

of xylem cell production connected to increasing cell volumes by water intake, there is only low need for carbon ([Cuny et al. 2015](#)). On the contrary, the following phase of cell wall thickening requires high carbon supply ([Verbancic et al. 2018](#)). In the beginning of the growing season, a majority of cells are in the elongation phase. Later on, the number of cells in the phase of cell wall thickening increase.

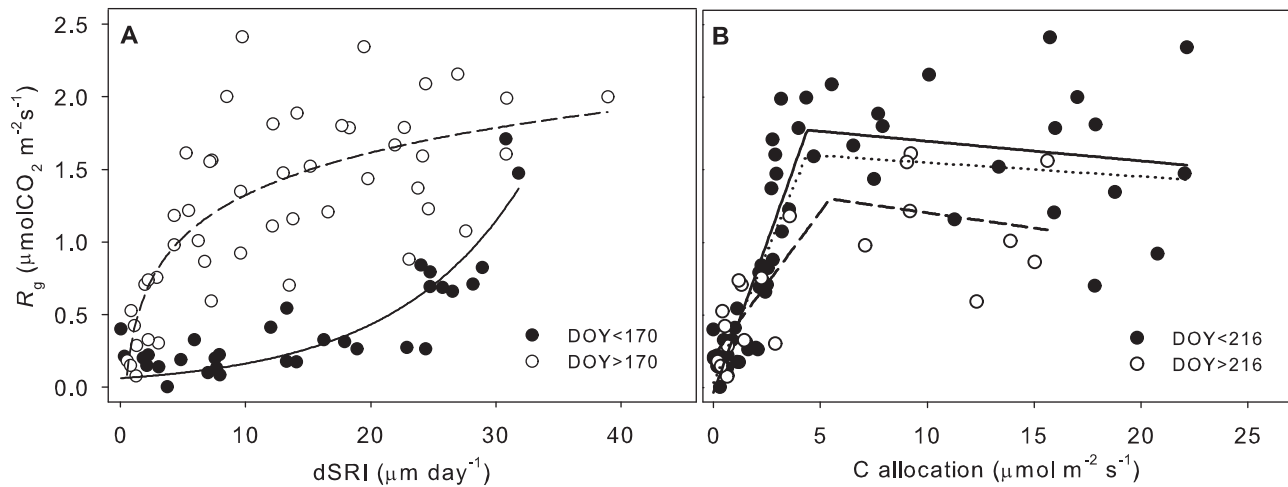


Figure 5. Relationship between stem growth respiration (R_g) and daily stem radial increment (dSRI) (A) and R_g and carbon allocation (C allocation) (B). The relationship between R_g and dSRI was divided before (solid line) and after (dashed line) day of year (DOY) 170 when the maximum dSRI was determined. The relationship between R_g and carbon allocation was assessed for all data (dotted line) and separately for periods before (solid line) and after (dashed line) DOY 216 when carbon allocation was the highest. The means from all studied trees are shown. Each dot represents day of stem radial increment measurement, corresponding to R_g obtained from 7-day running average and actual carbon allocation rate.

Moreover, cells produced in the first half of growing season have bigger volume but much thinner cell walls compared with the cells produced in the second half of growing season (Higuchi 1997). This supports our results when we observed the largest stem volumetric increment in the first half of the growing season but highest building material need (carbon allocation) in the second half of growing season (Figure 3).

Separation of R_m and R_g

The mature tissue method has been used in a number of studies (Ceschia et al. 2002, Zha et al. 2004, Wieser et al. 2009, Brito et al. 2010, Chan et al. 2018) because of its simplicity. There are, however, two issues concerning this method that have been debated. First, the assumption that maintenance sensitivity to temperature is constant over the year can be questioned. According to Amthor (1994), the short-term response of R_m to temperature measured in winter may not reflect the actual temperature response of R_m during summer. This can be caused by plant acclimation to changes in temperature (Linder and Troeng 1981, Paembonan et al. 1991). Because plants acclimated to higher temperatures have lower respiration rates than do plants acclimated to low temperatures when measured at the same temperature, the use of dormant-season rates to represent seasonal base respiration is likely to overestimate maintenance respiration during the warmer period. It is, however, difficult to assess seasonal temperature acclimation of R_m because R_m and R_g occur simultaneously in growing stems.

Maier (2001) also pointed out that R_m can differ between two consecutive postgrowing seasons, which he attributed to a relatively large change in sapwood volume. That study was done on fast-growing, young pine trees, while our study was done on

spruce trees. Sapwood area of Norway spruce trees is formed by several of the last peripheral tree rings. While new sapwood grows over the outer circumference, it turns into heartwood from the inner circumference. This results in the width of sapwood remaining more or less constant after reaching a certain age (Sellin 1994). Therefore, although stem diameter of our studied trees gradually increased from 2005 until 2013, we assume the width of sapwood under the respiration chamber to be more or less constant over the whole study period. We consider this width to be about 5 cm as determined by Pokorny (unpublished data) at the same site.

Stem respiration

Stem respiration (R_{st}) showed a distinct pattern in all years, with maxima occurring in summer months as is consistent with most other studies. The summer maxima have been attributed both to the highest temperature of the year and the energy demands from stem growth processes (Ceschia et al. 2002, Zha et al. 2004, Maier et al. 2010, Darenova et al. 2018). This corresponds also to our conditions.

As stated in the study by Darenova et al. (2018), stem temperature was a good predictor of R_{st} of the experimental trees, but the temperature sensitivity of R_{st} can be overestimated as a consequence of the stem growth. As stem growth followed the increase in temperature at the beginning of the growing season, R_{st} seemed to be more sensitive to temperature than it really was. That is in agreement also with Ceschia et al. (2002). Several studies have described lower temperature sensitivity of stem CO_2 efflux (Q_{10}) in summer (Lavigne and Ryan 1997, Stockfors and Linder 1998, Maier 2001, Zha et al. 2004). The study of Darenova et al. (2018), which is based on the same data set as is this study, showed the lowest Q_{10}

in July and August and the highest in October (a postgrowing period). One of the reasons for such seasonality of Q_{10} can be attributed to temperature acclimation of respiration (Linder and Troeng 1981), which leads to lower sensitivity of R_{st} in plants growing under warm conditions. Another reason, however, can be that the highest proportion of R_g usually occurs during these months (Figure 4). R_g has been assumed to be less sensitive or insensitive to changes in temperature (Adu-Bredu et al. 1997, Frantz et al. 2004) and is attributed only to growth.

We found R_g to contribute to 43–59% of total stem CO_2 efflux during the growing period from May to September. When extrapolating R_m over the postgrowing period, during which time R_{st} was not measured, the contribution of R_g decreased to 37–52%. Ryan (1990) and Stockfors and Linder (1998) found similar R_g contribution for the growing season (38–47%) in spruce trees of similar size to ours. On an annual basis, our results were at the upper end of the range between 12 and 54% reported for conifers (Lavigne et al. 1996, Lavigne and Ryan 1997, Stockfors and Linder 1998, Zha et al. 2004, Maier 2001, Brito et al. 2010). This range of R_g contribution to R_{st} is wide and can vary in dependence on various factors. For example, Ryan (1990) found lower contribution of R_g for trees with stems 12 cm in diameter compared with those with stems 36 cm in diameter, thus showing that the ratio of R_m and R_g differs between trees with different size, age and growth rate. Brito et al. (2010) attributed low annual R_g (16%) in their study to an extensive warm and dry period without growth, which suggests that conditions during the dormancy period can play an important role. Both R_m and R_g can be also species-specific, as demonstrated by Lavigne and Ryan (1997) who found higher R_g contribution to R_{st} in spruce trees compared with pine trees. The high contribution of R_g in our study may be attributed to the fast-growing Norway spruce species, stand age when trees intensively grow and suitable climatic conditions for growing.

Seasons 2006 and 2013 were characterized by very dry July. We assume this to be the reason for the rapid decrease in R_{st} and both its components in these periods. According to our results, however, the response of R_g to drought was sooner and greater than that of R_m , and that may indicate greater sensitivity of R_g to water deficit compared with R_m . The negative effect of drought on R_{st} has been observed in previous studies (Saveyn et al. 2007, Rodríguez-Calcerrada et al. 2014). The effect of drought on stem growth may differ depending upon its timing. Lack of water at the beginning of the growing season may be reflected in smaller numbers and volumes of cells and, therefore, smaller stem radial increment (Gruber et al. 2010, Pacheco et al. 2016). On the other hand, we may expect that later drought can affect cell wall thickening processes. Although Hacura et al. (2015) reported that final stem wall thickness in Norway spruce was not affected by drought, we assume that cell development processes may be slowed as a result of

drought-induced reduction of photosynthetic assimilate substrate. This can be supported by the study of Chan et al. (2018), who found that R_g followed changes in gross primary production. That production is significantly reduced during dry periods at our studied forest, as confirmed by Krupková et al. (2019). Less intensive cell wall thickening on the one hand but longer duration of this phase on the other can result in no final effect on cell wall thickness.

Stem respiration vs stem increment rate and carbon allocation

Stem growth results in an increased need of energy, and this need should correspond to the intensity of growth. Studies like Zha et al. (2004), Rodríguez-Calcerrada et al. (2014) or Araki et al. (2015) found a correlation between stem respiration and stem increment. On the contrary, no correlation was confirmed by Gruber et al. (2009). Evaluating this relationship for the whole growing season can be problematic as Chan et al. (2018) observed that the correlations between stem increment and respiration varied over the time. They found no correlation in the beginning of the growing season, while the highest correlations were in the summer and weaker correlations when growth slowed down. In our study, we also observed the relationship between R_g and daily stem increment; however, this relationship varied over the growing season.

Both later increase in R_g at the beginning and its later decline in the second half of the growing season compared with stem radial increment (Figure 5) may be attributed to different matter and energy demands in time. Individual phases of cell development have different energy and matter demands from the viewpoint of synthesizing pectins, structural saccharides (celluloses, hemicelluloses) and polyphenols (lignin) (Fengel and Wegener 1989). Use of compounds for production of new cells and building of primary cell walls is relatively small, because primary cell wall thickness ranges around 10^{-1} μm (Panshin and De Zeeuw 1980). On the contrary, cell wall thickening requires high energy and material supply (the secondary cell wall thickness ranges around 10^{-0} μm) (Panshin and De Zeeuw 1980).

The time lag of R_g behind radial stem increment may be related to cambium cell division and duration of the individual phases of new xylem cell development. In the beginning of the growing season (May), the cell division rate is low and the duration of phases long (Horáček et al. 1999, Cuny and Rathgeber 2016). Therefore, the majority of cells remain most of the time in the elongation phase (when only primary cell wall is built). Subsequently, increasing temperature stimulates enzymatic activity, cambium cell division accelerates, the cell period of differentiation gets shorter, and the cells turn more quickly to the phase of secondary thickening. As the energy for these processes is supplied from growth respiration, we can hypothesize that if the number of cells in the phase

of elongation exceeds the number of cells in the phase of differentiation, then respiration is low. If the opposite is true, then respiration is high despite the equal stem increment rate. At the end of the growing season (September, October), we would expect growth respiration resulting only from supply of energy for cell wall thickening, due to cessation of cambium cell division and of cell elongation (no stem radial increment is recorded). This lag of R_g behind stem increment is supported by the later onset of SCA xylem cell walls, which was the highest in July and August (Figure 2). The SCA maxima were, however, after maxima of growth respiration.

The respiration needed (or amount of released carbon) for carbon incorporation in a certain amount of structural matter (biomass), the so called growth respiration coefficient (Lavigne and Ryan 1997, Rodríguez-Calcerrada et al. 2019), has been used to estimate seasonal course of growth respiration. This coefficient was determined from yearly data and was supposed to be constant over the year. Ryan et al. (2009) found linear relationship between wood CO_2 efflux ($\mu\text{mol C mol C}^{-1} \text{ s}^{-1}$) and relative growth ($\mu\text{mol C mol C}^{-1}$), which was estimated from stem volumetric changes and wood density of harvested trees. We, however, assume that wood density varies over the season in dependence on the ratio of cells in the elongation and cell wall thickening phases and on whether early or late wood is developed, as the latter can have higher wood density because of smaller cells but thicker cell wall as observed for example by Fries and Ericsson (2009). These results are, however, inconsistent with our results. We observed a tight relationship between R_g and carbon allocation only up to carbon allocation of $4.4 \mu\text{mol m}^{-2} \text{ s}^{-1}$, and then the relationship was weak (Figure 5). From our data, we are not able to explain why no correlation between R_g and SCA above the threshold was determined. There might be some inaccuracy in carbon allocation estimation as that was determined on the basis of data from 2014 to 2017. However, Gričar et al. (2015) found that inter-annual variation in xylem anatomical parameters was very small, especially in climatic conditions favorable for tree growth. Some anomaly could be found in 2013 when a long dry period in July and August could affect carbon allocation to late wood, which is sensitive to climatic conditions. There is also a hypothesis that some respiration may occur without benefit to a plant and cannot be included either in R_m or in R_g (Amthor 2000). Such respiration is called wastage respiration and is induced by increased substrate supply occurring when sugar levels have been raised. We may speculate that intensive photosynthesis during summer might increase sugar concentration in the tree and contribute to the discrepancies in the relationship between stem growth and stem respiration.

To reveal the connection of respiration, CO_2 flux and energy demands with individual processes taking place in tree stems would require simultaneous measurements of stem

respiration and frequent periodic wood micro-core sampling for evaluating cell development phases accompanied by carbon content analyses. This might be an inspiration for future research.

Stem CO_2 efflux in a future climate

Yearly patterns and the ratio of R_m and R_g described in this and also previous studies may be expected to be modified by future climate changes. Temperature is one of the driving factors of enzymatic processes, and it will rise by as much as 2–4 °C in the Czech Republic before the end of the century (Dubrovsky et al. 2005). According to Zha et al. (2004), higher temperatures affect R_m rather than R_g . On the other hand, previous studies have shown that in uplands and mountain areas common for Norway spruce cultivation, tree growth, together with other phenological characteristics, will start earlier and the growing season will be extended into the late autumn (Dose and Menzel 2004, Koch et al. 2005). Such a prolongation of the growing season may increase the yearly cost for growth and, therefore, R_g . Moreover, in our study, we observed a decrease in stem respiration during a dry period and possible higher sensitivity of R_g to drought compared with R_m . The climate change will bring such stress periods, such as summer heat waves with tropical temperatures accompanied by low air humidity (Dubrovsky et al. 2005), more frequently, which will affect carbon flux in the stems in the long term.

Conclusions

Our study demonstrated that stem growth respiration R_g generally increased with dSRI, but the relationship between these two variables differed before and after DOY 170, the day when dSRI culminated. Before this date, the relationship was described with an exponential function, while after this date, it was described with a logarithmic function. We consider this to be a result of different proportions of cells in the elongation phase versus the phase of cell wall thickening. The latter is not reflected in stem increment, but we assume it to be more energetically demanding than is cell elongation. Although measurement of stem increment by dendrometers is a powerful tool for studying stem growth, it cannot record all its aspects. We, however, did not confirm our hypothesis that growth respiration would be the highest during the peak of the rate of carbon allocation to stem. That peak occurred in the late summer when growth respiration already tended to decrease. Therefore, examining the relationship between R_g , stem growth and carbon allocation would require additional research based on analyses of stem wood formation from periodic micro-coring. Such work could analyze both cell elongation and consecutive cell wall thickening and their dynamics in relation to climatic conditions.

Supplementary Data

Supplementary Data for this article are available at Tree Physiology Online.

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Authors' contributions

E.D. provided measurements of stem CO₂ efflux, analyzed efflux data and wrote the manuscript with input from all authors. P.H. analyzed the stem increment data and related them to stem CO₂ efflux. J.K. analyzed carbon allocation to stem. R.P. related the results to wood formation and climate change. M.P. designed the stem CO₂ efflux system and did the measurements.

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References

- Abbramoff MD, Magalhães PJ, Ram SJ (2005) Image processing with ImageJ part II. *Biophotonics Int* 11:36–43.
- Acosta M, Pavelka M, Pokorný R, Janous D, Marek MV (2008) Seasonal variation in CO₂ efflux of stems and branches of Norway spruce trees. *Ann Bot* 101:469–477.
- Adu-Bredu S, Yokota T, Hagihara A (1997) Temperature effect on maintenance and growth respiration coefficients of young, field-grown hinoki cypress (*Chamaecyparis obtusa*). *Ecol Res* 12:357–362.
- Amthor JS (1989) *Respiration and crop productivity*. Springer-Verlag, New York, NY, p 215.
- Amthor JS (1994) Plant respiratory responses to the environment and their effects on the carbon balance. In: Wilkinson RE (ed) *Plant–environment interactions*. Marcel Dekker, New York, NY, pp 501–554.
- Amthor JS (2000) The McCree-de Wit-penning de Vries-Thornley respiration paradigms: 30 years later. *Ann Bot* 86:1–20.
- Araki MG, Kajimoto T, Han Q, Kawasaki T, Utsugi H, Gyokusen K, Chiba Y (2015) Effect of stem radial growth on seasonal and spatial variations in stem CO₂ efflux of *Chamaecyparis obtusa*. *Trees Struct Funct* 29:499–514.
- Bolstad PV, Davis KJ, Martin J, Cook BD, Wang W (2004) Component and whole-system respiration fluxes in northern deciduous forests. *Tree Physiol* 24:493–504.
- Brito P, Morales D, Wieser G, Jimenez MS (2010) Spatial and seasonal variations in stem CO₂ efflux of *Pinus canariensis* at their upper distribution limit. *Trees* 24:523–531.
- Ceschia E, Damesin C, Lebaube S, Pontailler JY, Dufrene E (2002) Spatial and seasonal variations in stem respiration of beech trees (*Fagus sylvatica*). *Ann For Sci* 59:801–812.
- Chalupa V (1969) Počátek, trvání a ukončení vegetační činnosti u lesních dřevin. In: *Práce VÚLHM sv. 37. VÚLHM, Zbraslav-Strnady, Czech Republic*, pp 41–68.
- Chan T, Berninger F, Kolari P, Nikinmaa E, Holtta T (2018) Linking stem growth respiration to the seasonal course of stem growth and GPP of scots pine. *Tree Physiol* 38:1356–1370.
- Cuny HE, Rathgeber CBK (2016) Xylogenesis: coniferous trees of temperate forests are listening to the climate tale during the growing season but only remember the last words! *Plant Physiol* 171:306–317.
- Cuny HE, Rathgeber CBK, Lebourgeois F, Fortin M, Fournier M (2012) Life strategies in intraannual dynamics of wood formation: example of three species in a temperate forest in Northeast France. *Tree Physiol* 32:612–625.
- Cuny HE, Rathgeber CBK, Frank D, Fonti P, Fournier M (2014) Kinetics of tracheid development explain conifer tree-ring structure. *New Phytol* 203:1231–1241.
- Cuny HE, Rathgeber CBK, Frank D et al. (2015) Woody biomass production lags stem-girth increase by over one month in coniferous forests. *Nat Plants* 1:15160.
- Darenova E, Acosta M, Pokorný R, Pavelka M (2018) Variability in temperature dependence of stem CO₂ efflux from Norway spruce trees. *Tree Physiol* 38:1333–1344.
- Darenova E, Szatniewska J, Acosta M, Pavelka M (2019) Variability of stem CO₂ efflux response to temperature over the diel period. *Tree Physiol* 39:877–887.
- De Swaef T, De Schepper V, Vandegehuchte MW, Steppe K (2015) Stem diameter variations as a versatile research tool in ecophysiology. *Tree Phys* 35:1047–1061.
- Dinwoodie JM (1981) *Timber: its structure, nature and behaviour*. Van Nostrand Reinholds Co., Ltd, London, UK.
- Dose V, Menzel A (2004) Bayesian analysis of climate change impacts in phenology. *Global Change Biol* 10:259–272.
- Dubrovsky M, Nemesova I, Kalvova J (2005) Uncertainties in climate change scenarios for the Czech Republic. *Clim Res* 29:139–156.
- Fengel D, Wegener G (1989) *Wood: chemistry, ultrastructure, reactions*. Walter de Gruyter, Berlin, pp 613.
- Frantz JM, Cometti NN, Bugbee B (2004) Night temperature has a minimal effect on respiration and growth in rapidly growing plants. *Ann Bot* 94:155–166.
- Fries A, Ericsson T (2009) Genetic parameters for earlywood and latewood densities and development with increasing age in scots pine. *Ann For Sci* 66: article number 404.
- Gaumont-Guay D, Black TA, Griffis TJ, Barr AG, Morgenstern K, Jassal RS, Nesic Z (2006) Influence of temperature and drought on seasonal and interannual variations of soil, bole and ecosystem respiration in a boreal aspen stand. *Agric For Meteorol* 140:203–219.
- Giagli K, Gričar J, Vavrčik H, Gryc V (2016) Nine-year monitoring of cambial seasonality and cell production in Norway spruce. *iForest* 9:375–382.
- Gričar J, Prislan P, de Luis M, Gryc V, Hacurova J, Vavrčik H, Cufar K (2015) Plasticity in variation of xylem and phloem cell characteristics of Norway spruce under different local conditions. *Front Plant Sci* 6:730.
- Gričar J, Zupančič M, Cufar K, Oven P (2007) Regular cambial activity and xylem and phloem formation in locally heated and cooled stem portions of Norway spruce. *Wood Sci Technol* 41:463–475.
- Gruber A, Wieser G, Oberhuber W (2009) Intra-annual dynamics of stem CO₂ efflux in relation to cambial activity and xylem development in *Pinus cembra*. *Tree Physiol* 29:641–649.

- Gruber A, Stroh S, Veit B, Oberhuber W (2010) Impact of drought on the temporal dynamics of wood formation in *Pinus sylvestris*. *Tree Physiol* 30:490–501.
- Hacura J, Gryc V, Vavrick H, Hozova J, Urban J (2015) The effect of drought on cell wall thickness and radial dimension of tracheids of *Picea abies* (L.) Karst. *Wood Res* 60:175–187.
- Hannerz M (1999) Evaluation of temperature models for predicting bud burst in Norway spruce. *Can J For Res* 29:9–19.
- Higuchi T (1997) Formation of earlywood, latewood, and heartwood. In: *Biochemistry and molecular biology of wood*. Springer Series in Wood Science. Springer, Berlin, Heidelberg.
- Hölttä T, Mäkinen H, Nöjd P, Mäkelä A, Nikinmaa E (2010) A physiological model of softwood cambial growth. *Tree Physiol* 30:1235–1262.
- Horáček P, Šlezingerová J, Gandelová L (1999) Effects of environment on the xylogenesis of Norway spruce (*Picea abies* L. Karst.). In: Wimmer R, Vetter RE (eds) *Tree-ring analysis: biological, methodological and environmental aspects*. CABI Publishing, Wallingford, UK, pp 33–53.
- Khomik M, Arain MA, Brodeur JJ, Peichl M, Restrepo-Coupe N, McLaren JD (2010) Relative contributions of soil, foliar, and woody tissue respiration to total ecosystem respiration in four pine forests of different ages. *J Geophys Res* 115:G03024.
- Koch E, Bruns E, Chmielewski FM, Defila C, Lipa W, Menzel A (2005) Guidelines for plant phenological observations. World Meteorological Organization, WMO-TD No. 1484, Geneva, Switzerland.
- Kozłowski TT, Winget CH (1964) Diurnal and seasonal variation in radii of tree stems. *Ecology* 45:149–155.
- Krupková L, Havráňková K, Krejza J, Sedlák P, Marek MV (2019) Impact of water scarcity on spruce and beech forests. *J For Res* 30:899–909.
- Lavigne MB, Ryan MG (1997) Growth and maintenance respiration rates of aspen, black spruce and jack pine stems at northern and southern BOREAS sites. *Tree Physiol* 17:543–551.
- Lavigne MB et al. (1996) Estimating stem maintenance respiration rates of dissimilar balsam fir stands. *Tree Physiol* 16:687–695.
- Linder S, Troeng E (1981) The seasonal variation in stem and coarse root respiration of a 20-year-old scots pine (*Pinus sylvestris* L.). In: *Mitteilungen der forstlichen Bundesversuchsanstalt. Österreichischer Agrarverlag, Wien*, pp 125–139.
- Maier CA (2001) Stem growth and respiration in loblolly pine plantations differing in soil resource availability. *Tree Physiol* 21:1183–1193.
- Maier CA, Johnsen KH, Clinton BD, Ludovici KH (2010) Relationships between stem CO₂ efflux, substrate supply, and growth in young loblolly pine trees. *New Phytol* 185:502–513.
- Dose V, Menzel A (2004) Bayesian analysis of climate change impacts in phenology. *Global Change Biol* 10:259–272.
- Oberhuber W, Gruber A, Kofler W, Swidrak I (2014) Radial stem growth in response to microclimate and soil moisture in a drought-prone mixed coniferous forest at an inner alpine site. *Eur J For Res* 133:467–479.
- Pacheco A, Julio Camarero J, Carrer M (2016) Linking wood anatomy and xylogenesis allows pinpointing of climate and drought influences on growth of coexisting conifers in continental Mediterranean climate. *Tree Physiol* 36:502–512.
- Paembonan SA, Hagihara A, Hozumi K (1991) Long-term measurement of CO₂ release from the aboveground parts of a Hinoki forest tree in relation to air-temperature. *Tree Physiol* 8:399–405.
- Panshin AJ, De Zeeuw C (1980) *Textbook of wood technology*. McGraw-Hill Inc., New York, NY, p 723.
- Penman J, Gytarsky M, Hiraishi T et al. (eds). (2003) *Good practice guidance for land use. Forestry, land-use change and forestry*. IGES, Kanagawa, Japan.
- Ratgeber CBK, Decoux V, Leban J-M (2006) Linking intra-tree-ring wood density variations and tracheid anatomical characteristics in Douglas fir (*Pseudotsuga menziesii* (Mirb.) Franco). *Ann For Sci* 63:699–706.
- Rodriguez-Calcerrada J, Martin-StPaul NK, Lempereur M, Ourcival J-M, del Rey MDC, Joffre R, Rambal S (2014) Stem CO₂ efflux and its contribution to ecosystem CO₂ efflux decrease with drought in a Mediterranean forest stand. *Agric For Meteorol* 195:61–72.
- Rodriguez-Calcerrada J, Salomon RL, Gordaliza GG, Miranda JC, Miranda E, de la Riva EG, Gil L (2019) Respiratory costs of producing and maintaining stem biomass in eight co-occurring tree species. *Tree Phys* 39:1838–1854.
- Rossi S, Deslauriers A, Morin H (2003) Application of the Gompertz equation for the study of xylem cell development. *Dendrochronologia* 2:33–39.
- Rossi S, Anfodillo T, Menardi R (2006) Trephor: a new tool for sampling microcores from tree stems. *IAWA J* 27:89–97.
- Rossi S, Deslauriers A, Gricar J et al. (2008) Critical temperatures for xylogenesis in conifers of cold climates. *Glob Ecol Biogeogr* 17:696–707.
- Ryan MG (1990) Growth and maintenance respiration in stems of *Pinus contorta* and *Picea engelmannii*. *Can J For Res* 20:48–57.
- Ryan MG (1991) Effects of climate change on plant respiration. *Ecol Appl* 1:157–167.
- Ryan MG, Cavaleri MA, Almeida AC, Penchel R, Senock RS, Stape JL (2009) Wood CO₂ efflux and foliar respiration for eucalyptus in Hawaii and Brazil. *Tree Physiol* 29:1213–1222.
- Saveyn A, Steppe K, Lemeur R (2007) Drought and the diurnal patterns of stem CO₂ efflux and xylem CO₂ concentration in young oak (*Quercus robur*). *Tree Physiol* 27:365–374.
- Sellin A (1994) Sapwood heartwood proportion related to tree diameter, age, and growth-rate in *Picea abies*. *Can J For Res* 24:1022–1028.
- Skálová H, Moravcová L, Dixon AFG, Kindlmann P, Pyšek P (2015) Effect of temperature and nutrients on the growth and development of seedlings of an invasive plant. *AoB Plants* 7:plv044.
- Stockfors J, Linder S (1998) Effect of nitrogen on the seasonal course of growth and maintenance respiration in stems of Norway spruce trees. *Tree Physiol* 18:155–166.
- Tang J, Bolstad PV, Desai AR, Martin JG, Cook BD, Davis KJ, Carey EV (2008) Ecosystem respiration and its components in an old-growth forest in the Great Lakes region of the United States. *Agric For Meteorol* 148:171–185.
- Teskey RO, McGuire MA (2007) Measurement of stem respiration of sycamore (*Platanus occidentalis* L.) trees involves internal and external fluxes of CO₂ and possible transport of CO₂ from roots. *Plant Cell Environ* 30:570–579.
- Verbancic J, Lunn JE, Stitt M, Persson S (2018) Carbon supply and the regulation of cell wall synthesis. *Mol Plant* 11:75–94.
- von Arx G, Crivellaro A, Prendin AL, Čufar K, Carrer M (2016) Quantitative wood anatomy—practical guidelines. *Front Plant Sci* 7:1–13. doi: 10.3389/fpls.2016.00781.
- Wieser G, Gruber A, Bahn M, Catala E, Carrillo E, Soledad Jimenez M, Morales D (2009) Respiratory fluxes in a Canary Islands pine forest. *Tree Physiol* 29:457–466.
- Wodzicki TJ (1971) Mechanism of xylem differentiation in *Pinus sylvestris* L. *J Exp Bot* 22:670–687.
- Zha T, Xing Z, Wang K-Y, Kellomaki S, Barr AG (2007) Total and component carbon fluxes of a scots pine ecosystem from chamber measurements and eddy covariance. *Ann Bot* 99:345–353.
- Zha TS, Kellomaki Z, Wang KY, Ryyppo A, Niinisto S (2004) Seasonal and annual stem respiration of scots pine trees under boreal conditions. *Ann Bot* 94:889–896.
- Zweifel R, Item H, Hasler R (2001) Link between diurnal stem radius changes and tree water relations. *Tree Physiol* 21:869–877.